

Course Level: Undergraduate Open Elective (B. Tech)

BL: 2 Understand

CO: Recall the basics of food groups, food composition, nutritive value and functions of food.

Module 2: Macronutrients: Proteins

1. Overview

Proteins = building blocks of life.

Composed of C, H, N, O.

Natural polymers of amino acids → fold into globular or fibrous structures.

20 standard amino acids used to build thousands of proteins.

2. Formation

Peptide bond: covalent bond formed via dehydration (carboxyl group + amino group → water released).

Stable planar amide backbone.

3. Chemical Properties

Amphoteric nature: act as acids or bases → buffer capacity.

Isoelectric point (pI): pH where protein has zero net charge.

Colloidal behavior: large molecules form colloids in water, solubility depends on shape, amino acid composition, salt concentration, pH.

4. Classification

Simple proteins: yield only amino acids (e.g., albumin, collagen, keratin).

Conjugated proteins: contain prosthetic groups (glycoproteins, lipoproteins, nucleoproteins).

Derived proteins: formed by denaturation/degradation (coagulated proteins, proteoses, peptones).

5. Core Physiological Functions

- Enzymatic catalysis: nearly all metabolic reactions driven by enzymes (proteins).
- Growth & maintenance: structural matrix of bones, muscles, skin, hair, organs.
- Transport: hemoglobin (oxygen), lipoproteins (fats).
- Hormones & neurotransmitters: insulin, growth hormone, peptide messengers.
- Fluid balance: albumin maintains osmotic pressure, prevents edema.
- Acid-base balance: proteins buffer blood pH.
- Defense & detoxification: antibodies, liver enzymes.
- Energy source: 1 g protein = 4 kcal (used when carbs are depleted).

6. Dietary Sources & Processing

Complete proteins (animal sources) + complementary plant proteins ensure amino acid requirements.

Thermal processing:

Moderate heat → denaturation → improves digestibility.

Excessive heat → damages amino acids (e.g., lysine), reduces nutritional value, may form toxic compounds.

7. Protein Requirements & Nitrogen Balance

Requirement depends on growth stage & gender.

Nitrogen balance:

$N\text{ balance} = N\text{ intake} - (N\text{ losses in urine, feces, sweat}).$

Positive balance essential during growth, pregnancy, adolescence.

Average Protein Requirement (APR):

Below 6 months: 66% used for growth.

Above 6 months: 58% used for growth.

8. Deficiency Disorders

Kwashiorkor: protein deficiency despite adequate energy intake.

Marasmus: general malnutrition (deficiency of energy + protein).